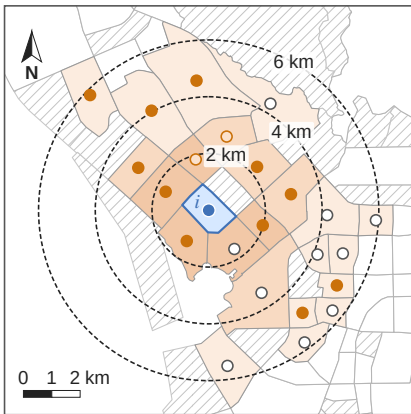
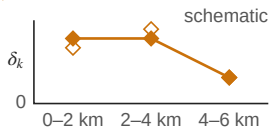


Distance-ring spillover exposure



- zone i (TOU, commercial)
- ring 1: $[0, 2)$ km, 5 zones
- ring 2: $[2, 4)$ km, 5 zones
- ring 3: $[4, 6)$ km, 13 zones
- zones beyond 6 km
- no charging data

- strict TOU
- weak TOU
- fixed price ($\Delta \ell_j \equiv 0$)
- ◇ raw $\hat{\delta}_k$ from switch events
- ◆ used: PAV to $\delta_1 \geq \delta_2 \geq \delta_3 \geq 0$



$S_{i,t}^{(k)} = \text{mean}\{\Delta \ell_{j,t} : j \in \mathcal{R}_k(i)\}$, $\mathcal{R}_k(i) = \{j \neq i : r_{k-1} \leq d_{ij} < r_k\}$, d_{ij} : centroid distance

$\eta_{i,t}^{\text{spill}} = \sum_k \delta_k S_{i,t}^{(k)}$; an empty ring gives $S_{i,t}^{(k)} = 0$